

COMMUNICATIONS LAB. Experiment #8: Octave Adaptation Experiment

Student ID / Name and Surname : Mert Can Çevik 151220192058

Student ID / Name and Surname : Mustafa Yağcı 151220192027

By turning this report, you agree that the work presented here are the results of your own experiment and the comments are your own.

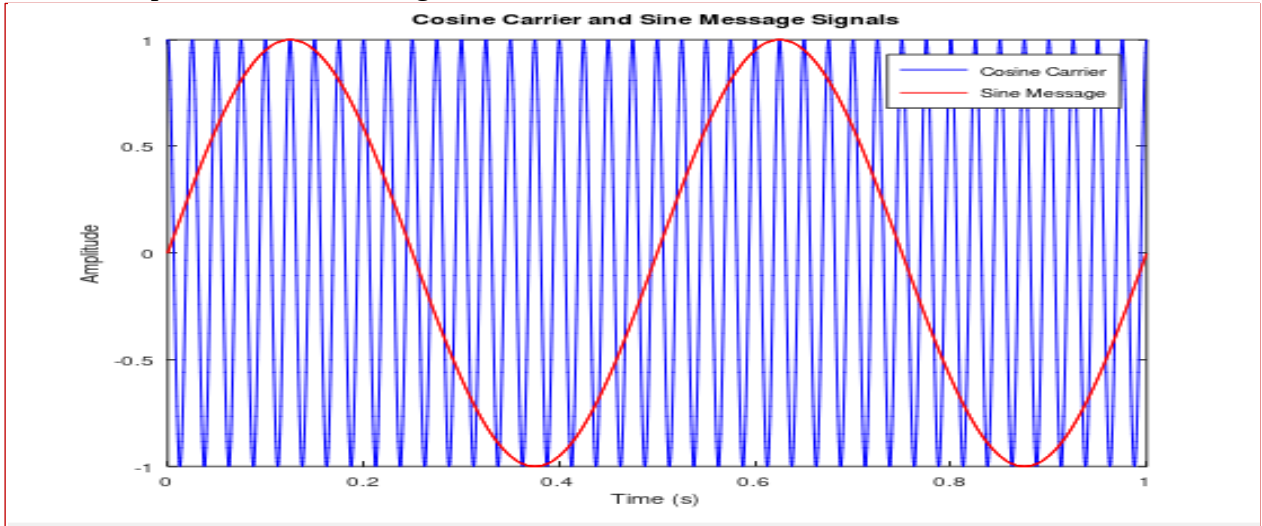
NOTES / COMMENTS / DRAWINGS FROM YOUR EXPERIMENT

Insert Octave plots within the drawing canvasses below.

b) Your expression for generating carrier signal samples :

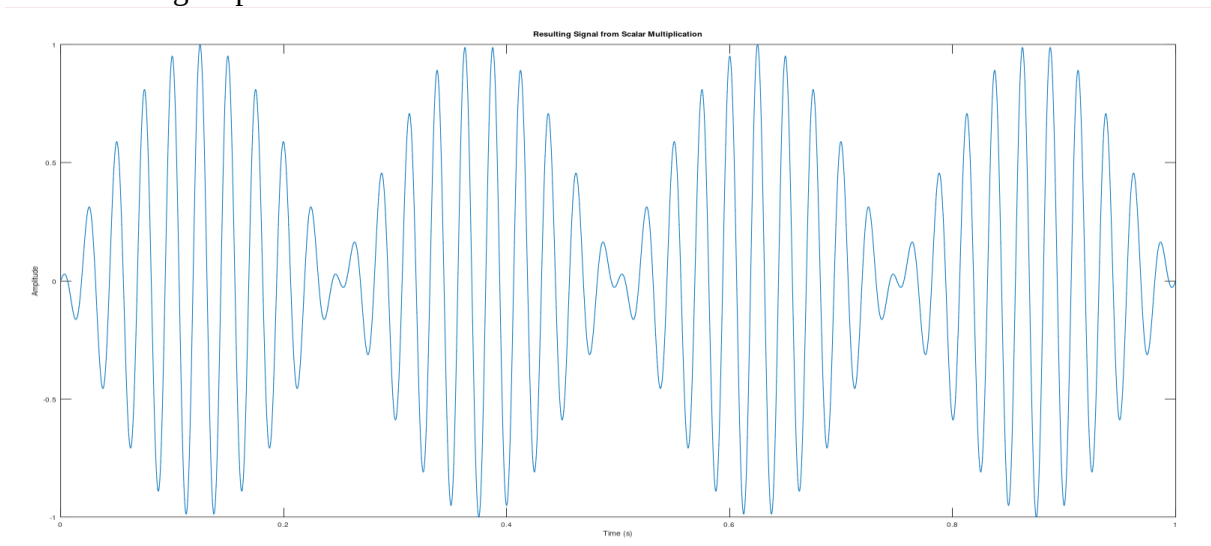
Your expression for generating a sine message signal :

Insert both plots into the drawing canvas below:



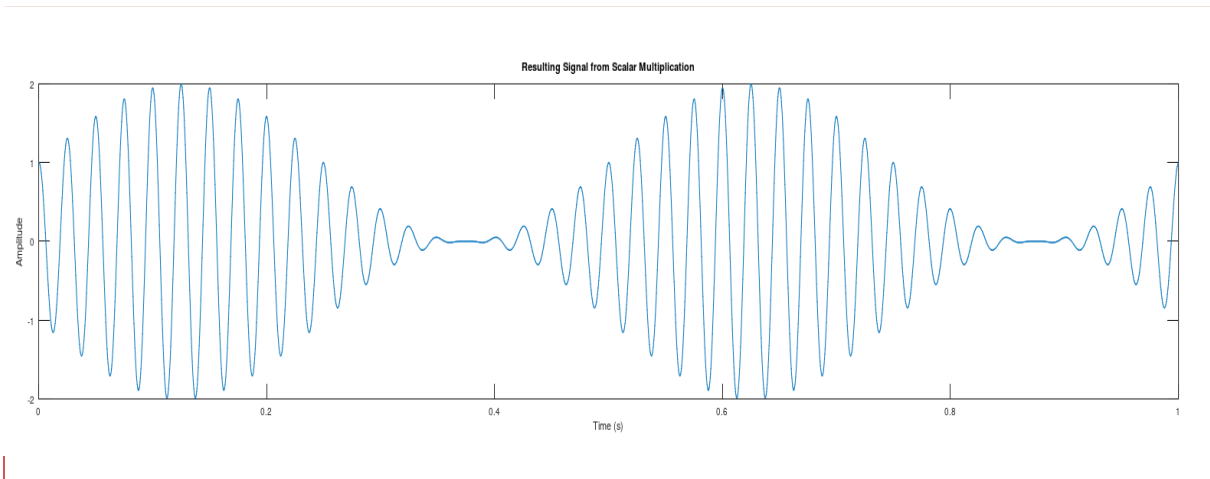
c) Expression for scalar multiplication of the carrier and message signal samples:

Modulated signal plot below:



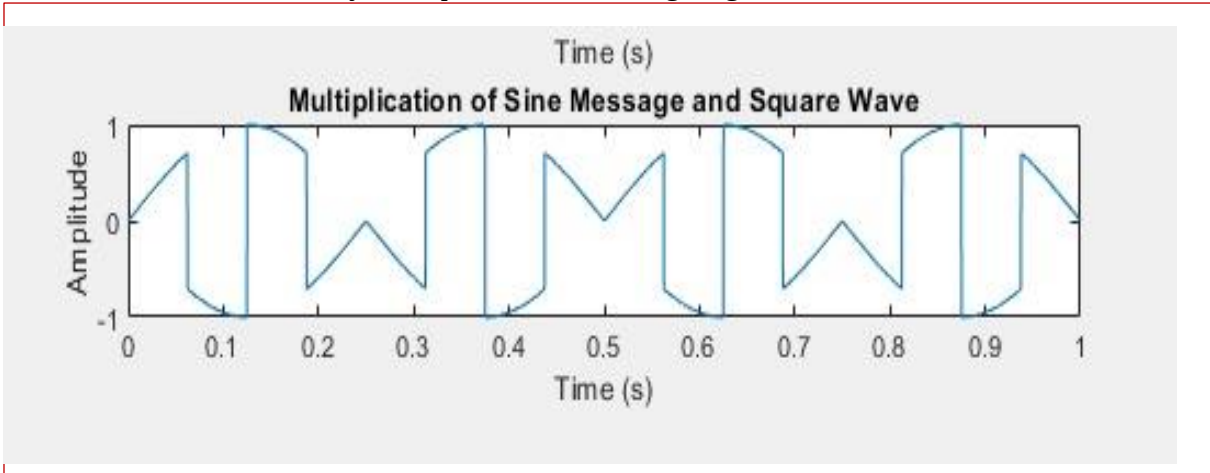
e) Expression for scalar multiplication of the carrier and message signal samples:

Modulated signal plot below:

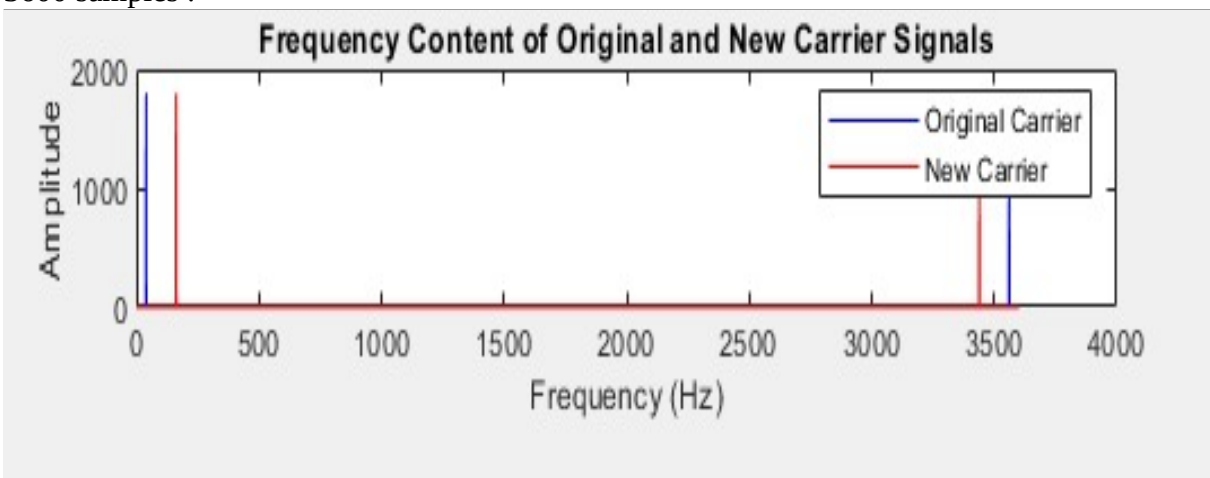


f) Expression(s) for generating square wave :

Plot of carrier modulated by the square wave message signal below:



g) FFT plots (on the same graph with different colors) of carriers with 4 and 4.05 periods within 3600 samples :



QUESTIONS

- 1- Why should we be careful while obtaining frequency spectrum using FFT? What should we be aware of?

When using FFT to get frequency spectra, we should be careful about aliasing due to insufficient sampling, spectral leakage from windowing, understand zero-padding's effect on resolution, and balance frequency - time resolution.

- 2- Do you think FFT can be used to obtain frequency spectrum of continuous signals? Explain.

No, FFT can't directly analyze continuous signals because it works on discrete-time data; first, continuous signals need to be sampled into discrete data for FFT analysis

CONCLUSION

Write down what you have learned with this experiment.

Through this experiment, we learned how to generate and manipulate signals for modulation, including amplitude modulation and the effects of DC biasing, as well as how different frequency components influence signal characteristics.